

Formalizing Brouwer's Fixed Point Theorem via Sperner's Lemma

MATH 157 Final Project: Essay

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1. Precise Statements

Throughout this project the ambient space is the standard 2-simplex

$$\Delta^2 = \{x : \text{Fin } 3 \rightarrow \mathbb{R} \mid (\forall i, 0 \leq x_i) \wedge \sum_i x_i = 1\}$$

equipped with the subspace topology inherited from $\mathbb{R}^3$. The n th barycentric subdivision of Δ^2 places a vertex at every point $(a/n, b/n, 1 - a/n - b/n)$ with $a, b \in \mathbb{N}$ and $a + b \leq n$. The set of such vertices is `TVertex n`. Each upward or downward triangle in the subdivision is recorded as a `STriangle n` value, whose three vertices are given by `STriangle.vert`.

Sperner's Lemma (2D).

Let $n \geq 1$. A Sperner labeling for the n th subdivision is a function $L : \text{TVertex } n \rightarrow \text{Fin } 3$ satisfying three corner conditions and one face condition:

$$\text{corner0: } L(0, 0) = 2 \quad \text{corner1: } L(n, 0) = 0 \quad \text{corner2: } L(0, n) = 1$$

face: if v lies on face k (the face opposite vertex k) then $L(v) \neq k$.

A triangle T is *fully labeled* if its three vertices receive all three labels 0, 1, 2. Sperner's Lemma asserts that the number of fully labeled triangles is odd, hence at least 1. In the formalization this is stated as:

```
axiom sperner_lemma (n : N) (hn : 0 < n)
  (L : SpernerLabeling n) (hL : IsSpernerLabeling n hn L) :
  Odd (Finset.card ((allTriangles n).filter (IsFullyLabeled n L)))
```

Brouwer's Fixed Point Theorem (2D).

Every continuous map $f : \Delta^2 \rightarrow \Delta^2$ has a fixed point. Formally:

```
theorem brouwer_fixed_point_2d (f : (Fin 3 -> R) -> Fin 3 -> R)
  (hf_cont : Continuous f)
  (hf_mem : forall x, x in stdSimplex2 -> f x in stdSimplex2) :
  exists x in stdSimplex2, f x = x
```

2. Proof Pathway

The proof connects a combinatorial parity argument (Sperner) to a topological fixed-point conclusion (Brouwer) through the following chain.

Step 1: Labeling from a fixed-point-free map.

Assume for contradiction that f has no fixed point, so $f(x) \neq x$ for all $x \in \Delta^2$. For any vertex v in the n th subdivision, the barycentric coordinate vector $p = \text{TVertex.toPoint } n \ v$ lies in Δ^2 and $f(p) \neq p$. Since both p and $f(p)$ lie in Δ^2 with equal coordinate sums, there must exist some coordinate index k where $f(p)_k < p_k$. We define `labelFromMap` by choosing the smallest such k via `Classical.choose`. The lemma `labelFromMap_is_sperner` then verifies all four conditions of `IsSpernerLabeling`. The corner conditions follow from the fact that at each corner vertex one barycentric coordinate equals 1 and the rest equal 0, so only one coordinate can decrease. The face conditions follow because on face k the k th coordinate is 0, so $f(p)_k \geq 0 = p_k$ and k is never the chosen index.

Step 2: Fully labeled triangles exist at every scale.

By `sperner_lemma` applied to the Sperner labeling from Step 1, the filter `(allTriangles n).filter (IsFullyLabeled n L.n)` has odd cardinality for every $n \geq 1$. An odd number is positive, so there exists at least one fully labeled triangle T_n at every scale.

Step 3: Convergence to a fixed point (proof sketch).

Let p_n be the barycenter of T_n . Since Δ^2 is compact (proved via `IsCompact.of_isClosed_subset`), the sequence (p_n) has a convergent subsequence $p_{\varphi(k)} \rightarrow x^*$ by `IsCompact.tendsto_subseq`. The diameter of any triangle in the n th subdivision is at most $2/n$, so all three vertices of $T_{\varphi(k)}$ also converge to x^* . For each coordinate index j , the vertex of $T_{\varphi(k)}$ with label j satisfies $f(v)_j < v_j$. Passing through the limit via `ContinuousAt.tendsto` and `ge_of_tendsto` gives $f(x^*)_j \leq x_j^*$ for each j . Summing over all j and using $\sum_j f(x^*)_j = \sum_j x_j^* = 1$ forces $f(x^*)_j = x_j^*$ for every j , contradicting the assumption that f has no fixed point. This step is documented as a `sorry` with the full proof sketch inline; the earlier steps are fully mechanized.

3. What Formalization Reveals

3.1 Well-definedness requires the fixed-point-free hypothesis.

In informal presentations the labeling “assign label k if coordinate k decreases under f ” is stated casually. Lean forces precision: `exists_coord_gt` must be proved before `Classical.choose` can be applied, and its proof uses both the simplex membership of $f(v)$ and the inequality $f(v) \neq v$. Without the fixed-point-free assumption there is simply no decreasing coordinate and the labeling is undefined. The formal proof makes this logical dependency explicit.

3.2 Corner conventions must be globally consistent.

The standard labeling assigns label k to corner e_k . In the formalization corner $e_0 = (1, 0, 0)$ is the vertex where `TVertex.1.val = n` and `TVertex.2.val = 0`; corner $e_1 = (0, 1, 0)$ is where `TVertex.2.val = n`; and corner $e_2 = (0, 0, 1)$ is the origin (both coordinates zero). The natural mnemonic “corner0 gets label 0” is therefore wrong: corner0 in the code is the origin, which is e_2 and should receive label 2. An early mismatch between `IsSpernerLabeling` and `labelFromMap` caused the proof of `labelFromMap_is_sperner` to fail silently. Lean’s type-checker surfaced this only when the full proof was attempted, illustrating how convention errors propagate invisibly until mechanization forces a check.

3.3 Finite cardinality requires non-trivial infrastructure.

Stating “the number of fully labeled triangles is odd” requires that `STriangle n` be a `Fintype`. This was not automatic: `STriangle n` is a structure with a `Prop`-valued validity field, which is not a plain subtype. The solution was to construct an explicit `Equiv` between `STriangle n` and the decidable subtype $\{p : \text{TVertex } n \times \text{Orient} \mid \text{valid } p\}$, using `Fintype.ofEquiv`. The `DecidablePred` instance for the validity condition required `cases o` (not `fin_cases`, which fails for `Prop`-valued targets) to reduce the match expression before `infer_instance` could synthesize the `Decidable` instance. These are the kinds of infrastructure gaps that paper proofs never expose.

3.4 The combinatorial and analytic parts are genuinely separate.

Treating `sperner_lemma` as an axiom is not merely a convenience. It reflects a real modularity in the proof: the Sperner argument is a finite combinatorial parity argument over a concrete triangulation, while the Brouwer argument is an analytic limit construction. The two parts use entirely different Mathlib infrastructure (`Finset` vs. `Metric.tendsto_subseq`), and keeping them separate makes each piece independently verifiable. The axiom is an honest statement of what remains to be mechanized rather than a structural hole.

3.5 Classical choice is unavoidable here.

The function `labelFromMap` is `noncomputable`: it uses `Classical.choose` to pick a decreasing coordinate. This is philosophically significant. The labeling exists and is well-defined, but there is no canonical choice of which decreasing coordinate to pick when multiple coordinates decrease. Lean’s `noncomputable` annotation makes this explicit and prevents the labeling from being mistakenly used in a decision procedure.